AI DRIVEN BEHAVIORAL AND MOTION BASED CONTROL OF ELECTRICAL APPLIANCES FOR PHYSICALLY CHALLENGED INDIVIDUALS

PROJECT EVALUATION II REPORT

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in

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DECLARATION

We affirm that the Project report titled "**AI Driven Behavioral And Motion Based Control Of Electrical Appliances For Physically Challenged Individuals** " being submitted in partial fulfillment for the award of **Bachelor of Technology** in **Computer Science and Business Systems**, is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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We gratefully remember our beloved **Founder Chairman, (Late) Thiru. M. Kumarasamy**, whose vision and legacy laid the foundation for our education and inspired us to successfully complete this project.

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Abstract (Key words)	POS Mapping
AI-driven Automation, Motion-based Control, Physically Challenged Assistance, Raspberry Pi, PIR Motion Sensors, Linear Regression Prediction, Smart Home Accessibility, Behavioral Analytics, Contactless Appliance Control, IoT-based Remote Monitoring, Assistive Technology, Energy-efficient Automation, Machine Learning Integration, User-Centered Smart Environment, Real-Time Appliance Control.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3

SDG Goals		Remarks
SDG 3	Enhances comfort, reduces dependency on caregivers, and supports safe living environments for physically challenged individuals.	This project demonstrates how emerging technologies can be applied responsibly to improve human life, especially for socially and physically disadvantaged groups. By combining AI, IoT, and motion sensing, the system offers smart, adaptive, and reliable.

ABSTRACT

Assistive automation plays a significant role in improving the independence, safety, and quality of life of physically challenged individuals. This project presents an AI driven behavioral and motion based control system for electrical appliances using an embedded platform. The system utilizes a PIR motion sensor to detect user presence and a current sensor to monitor appliance status. A Raspberry Pi is used as the central processing unit to collect and process sensor data in real time.

The proposed system implements a Linear Regression algorithm to analyze user behavior and predict appliance usage based on historical data such as time, motion detection, and previous usage patterns. The collected data is processed using Flask API and stored in MongoDB for continuous learning and analysis. The machine learning model enables the system to adapt to user habits and make intelligent decisions for automatic appliance control.

Experimental results show that the system achieves a motion detection accuracy of approximately 92% to 95% and a behavior prediction accuracy of around 85% to 90%. The system also demonstrates a fast response time of about 1 to 2 seconds, ensuring real time operation. Additionally, the automated control mechanism contributes to energy savings of nearly 20% to 30% compared to manual operation.

Keywords: Artificial Intelligence, Motion Detection, Behavioral Analysis, Raspberry Pi, Machine Learning, Linear Regression, PIR Sensor, Current Sensor, Embedded System, Smart Appliance Control, Assistive Technology, Real Time Processing, Energy Efficiency, MongoDB, Automation.

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LIST OF ABBREVIATIONS

S NO	ABBREVIATION	EXPANSION
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	PIR	Passive Infrared Sensor
4	WCS	Current Sensor
5	API	Application Programming Interface

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Individuals with physical disabilities often face significant challenges in performing daily activities that involve operating household electrical appliances requiring manual interaction. People with motor impairments, limited limb movement or neurological disorders may find it difficult or even impossible to switch on lights, operate fans, or control basic home devices. With advancements in assistive technology, the integration of artificial intelligence and embedded systems has enabled the development of intelligent, accessible and user centered solutions. These technologies help enhance independence and improve the overall quality of life for physically challenged individuals. Therefore, designing an intuitive and touch free system for controlling home appliances is an essential step toward creating inclusive smart environments.

Recent developments in computer vision, machine learning, and gesture recognition have opened new possibilities for designing systems that can accurately interpret user actions. By integrating these technologies with a Raspberry Pi, it is possible to develop a cost effective and flexible embedded system that supports real time processing, motion detection, and behavior analysis without requiring expensive hardware.

AI based systems allow users to control appliances through natural gestures, facial expressions, and small body movements, thereby reducing the dependency on physical interaction. This approach eliminates the need for manual switches and voice commands, making the system more suitable for individuals with mobility or speech limitations.

1.2 Overview of Assistive Home Automation Systems

For individuals with physical disabilities, interacting with standard home environments presents significant daily challenges. Assistive home automation systems are designed to bridge this gap by replacing manual, physical tasks with electronic alternatives. Historically, these systems have evolved from simple mechanical reach-extenders to remote controls and voice activated smart plugs. The current frontier of this technology involves edge computing and artificial intelligence, allowing the home environment to autonomously respond to the user's physical presence and behavior without requiring explicit commands.

Figure : 1.1 Overview Of Home Automation

1.3 Limitations of Traditional Electrical Switches

Despite advancements in smart homes, standard electrical infrastructures remain fundamentally inaccessible to those with limited mobility. The reliance on traditional mechanical switches presents several critical limitations:

1.3.1 Physical Accessibility Barriers

Standard electrical panels require reach, grip strength, and physical exertion. For wheelchair bound individuals or those with severe motor impairments, simply turning on a light or a fan can be impossible without caregiver assistance fundamentally limiting personal independence.

1.3.2 Lack of Adaptability to User Behavior

Traditional appliances, and even basic timer based smart plugs, do not adapt to the user. A timer set to turn a light off at 10:00 PM will do so regardless of whether the user is still reading or active in the room, creating frustrating and inflexible living environments.

1.3.3 Inefficient Energy Consumption

Because traditional switches are difficult to reach, physically challenged users often leave appliances (like fans or lights) running continuously throughout the day to avoid the strain of turning them off, leading to massive, unnecessary energy consumption.

1.4 Importance of Smart Automation in Assistive Care

The integration of smart automation into assistive care represents a paradigm shift from basic convenience to essential, life enhancing technology. For physically challenged individuals, relying on caregivers for simple environmental adjustments such as turning on a fan or switching off a light can lead to feelings of dependency and reduced self-esteem. Smart automation addresses this by restoring a critical sense of autonomy, privacy and personal dignity. It empowers users to manage their immediate surroundings independently, without requiring physical exertion or vocal strain.

An intelligent systems equipped with predictive AI go beyond the capabilities of simple remote controls; they proactively adapt to the user's daily routines, ensuring a safe and comfortable living space while preventing accidents (such as falls) that often occur when trying to reach mechanical switches. This level of automation also provides immense benefits to caregivers and family members. By autonomously managing routine environmental controls, caregivers are relieved of constant, repetitive monitoring, significantly reducing caregiver fatigue. Ultimately, smart automation transforms a restrictive physical

environment into an accessible, responsive and empowering space, drastically improving the overall psychological well being and quality of life for the user.

1.5 Role of IoT in Automation and Control

The Internet of Things (IoT) provides the critical infrastructure required to make assistive automation possible. By networking sensors and microcomputers, the physical environment becomes measurable and controllable.

1.5.1 Real Time Monitoring

Through the use of Passive Infrared (PIR) sensors and Real Time Clock (RTC) modules, IoT enables the continuous, real time monitoring of user occupancy and movement, ensuring the system always knows the state of the room.

1.5.2 Edge Processing and AI Control

Using edge devices like the Raspberry Pi, IoT systems can process data locally. Instead of relying on manual inputs, AI algorithms (like Linear Regression) process the IoT sensor data to learn routines and automatically actuate isolated relays, controlling high voltage appliances safely.

1.5.3 Energy Usage Analytics

IoT architectures equipped with current sensors can track exactly how much power is being drawn. This data is logged into CSV files and sent to cloud databases (like MongoDB Atlas), allowing the AI to analyze usage trends and optimize power efficiency.

1.6 Research Objectives

The specific objectives of this project are:

1. To design a touchless automation framework utilizing a Raspberry Pi, PIR sensors, and relay modules.
2. To implement a Python-based machine learning model that learns daily user habits from historical CSV logs.
3. To develop a dual verification control logic that combines predicted schedules with real-time motion data to actuate appliances.
4. To integrate an edge to cloud architecture using Node.js and MongoDB Atlas for continuous data synchronization and model updating.

1.7 PROBLEM STATEMENT

People with physical difficulties can find it hard to complete everyday tasks that require switches for turning on lights, fans, and other electrical devices at home. Traditional switch interfaces that use remote controls wall switches or voice commands are often not accessible to those who have limited use of their hands, have difficulties with speaking or who have neurological disorders. Therefore, this has resulted in those individuals having less independence, increasing their reliance on others for assistance and limiting their ability to participate in home activities daily. The lack of intuitive, adaptive and touchless control mechanisms is a significant impediment to helping these individuals live.

The current assistive technologies that provide control over devices primarily use mechanical switch activation, pressure sensors, or voice operation. These types of technologies have a number of drawbacks. Voice operated devices are less reliable due to problems from background noise, differences in dialects or individuals who have a speech disorder. Basic gesture recognition systems that use infrared sensors or simple image processing are also unreliable because they cannot accommodate for differences in ambient light, the presence of objects that are in the user’s way, or individual differences in movement for the same gesture.

There are also significant challenges based on the inability to accurately identify fine gesture and behavioural cues of individuals who demonstrate slow voluntary or involuntary, limited, or no motion at all. The conventional vision and movement recognition systems suffer from these issues when attempting to identify the users' gestures which may vary in terms of their orientation, speed and clarity caused by environmental considerations. Thus, there is a need for the development of a more sophisticated AI based framework for the identification of the more complex, highly variable and unpredictable human motion patterns under real-time conditions.

In short, the real world challenge of creating a behavioral and movement based appliance control system provides a true opportunity to support persons with physical disabilities by providing the means to utilize appliances without needing to use their hands or need to interact with anything physically via contactless operation. The overall goal of creating an Intelligent, Real Time control framework will provide a higher level of accessibility, independence, and overall improved quality of life for individuals with physical disabilities than could ever be achieved if the means of providing these controls did not include touchless and adaptive methods for appliance control. Currently, assistive technologies for appliance controls rely on mechanical switches, pressure sensors, or voice recognition based systems.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The development of assistive technologies for physically challenged individuals has gained considerable attention in recent years, driven by advancements in artificial intelligence, IoT and embedded systems. Traditional household appliances are primarily designed with manual interfaces, which pose accessibility challenges for individuals with limited motor skills. To address this numerous research initiatives have focused on incorporating intelligent automation to create more inclusive living environments.

Early systems primarily relied on mechanical switches or basic sensor based assistive controls. These approaches lacked adaptability and required some level of physical effort or close proximity to the device. Later improvements introduced voice-activated systems which provided hands free operation but were limited by speech impairments, noisy environments, and language barriers. Gesture based control methods also emerged, utilizing infrared or image based techniques. While these systems offered contactless interaction, they often struggled with variations in lighting, background conditions and user specific mobility constraints.

Recent studies have shifted toward AI driven solutions, where machine learning algorithms are employed to learn user behavior and predict appliance usage patterns. With the integration of IoT and cloud technology, remote monitoring and real time decision making have become increasingly feasible. These advanced systems not only focus on accessibility but also emphasize energy efficient automation, safety, and adaptability to diverse disability conditions.

2.2 AI Based Gesture and Motion Recognition for Assistive Control

Recent advancements have heavily focused on touchless appliance operation. Hobbs and Ali (2024) developed an AI based control system utilizing a Raspberry Pi and camera modules to process video streams. By extracting critical gesture points using lightweight computer vision algorithms, the system accurately translates user gestures into control commands for relays, offering a user friendly alternative to mechanical switches. Similarly, Chattoraj, Vishwakarma and Paul (2025) introduced a gesture classification system using lightweight Convolutional Neural Networks (CNNs). Their system employs adaptive threshold values to distinguish between intentional and unintentional movements, ensuring reliable operation even in complex backgrounds.

2.3 Deep Learning for Behavior Based Smart Home Automation

Deep learning frameworks are increasingly used to monitor more subtle user cues. Hussein and Adda (2025) proposed a smart home automation system that observes facial cues, hand movements and upper torso behavior. By employing convolution based gesture detection alongside context aware behavioral algorithms, the system successfully distinguishes non compliance movements from intentional commands, significantly increasing independence for disabled users.

2.4 Vision Assisted Human Motion Tracking

Tracking complex human motion requires robust image processing. Swami (2025) explored a vision assisted tracking system that converts movements such as head position and arm motions into command inputs. By utilizing techniques like background subtraction, key point extraction, and noise filtering on a Raspberry Pi edge device, the system maintains high reliability across varying lighting scenarios and physical backgrounds.

2.5 Adaptive Machine Learning Using User Motion Patterns

Static, rule based systems often fail to account for the natural variability of human movement. Kabir et al. (2024) addressed this by developing an adaptive machine learning platform that captures an individual's movement characteristics over time. As the user interacts with the system, the model continuously amends the motion profile, refining gesture recognition speed and noise filtering without requiring manual recalibration. This creates a frictionless and highly customized experience.

2.6 IoT Based Predictive Control of Home Appliances

Moving beyond real-time gesture recognition, predictive control anticipates user needs. Oladele and Markus (2024) introduced a system that combines infrared sensor data, current measurements, and Real Time Clock (RTC) timestamps to independently operate household devices. By processing daily CSV logs the AI model generates automated task schedules, minimizing unnecessary appliance activation and maximizing energy conservation.

2.7 Smart Behavior Learning and Context Aware Frameworks

Building upon predictive IoT control, Prasad et al. (2025) developed an Energy Optimization Strategy that dynamically adjusts activation schedules based on behavioral learning algorithms. The system includes fallback mechanisms for abnormal sensor readings. Validating this approach, Hamza et al. (2023) designed a context aware predictive framework using PIR networks and current sensors. Their tests confirmed that context-aware prediction provides a smoother operational experience and higher accuracy than simple timer based automation.

2.8 Comparison of Existing Assistive Control Models

While traditional systems rely heavily on rigid rule definitions or mechanical switches, modern models prioritize adaptability. Vision based models (Hobbs & Ali; Swami) excel in direct, intentional control but can be computationally heavy. Predictive IoT models (Oladele & Markus; Hamza et al.) are highly energy efficient and reduce physical strain by anticipating needs, but require consistent user routines. Advanced control frameworks (Arivarasan & Rahul) bridge these gaps by ensuring that regardless of the input method, the physical hardware actuates safely and smoothly without erratic behavior.

2.9 Summary

The literature survey highlights a crucial transition from simple timer based and mechanical switch controls to dynamic, AI driven environments. By integrating predictive behavioral models, touchless motion tracking, and robust nonlinear control strategies on edge devices like the Raspberry Pi, it is highly feasible to construct an assistive smart home system that is safe, adaptive, and significantly improves the quality of life for physically challenged individuals.

CHAPTER 3
PROPOSED SYSTEM

3.1 INTRODUCTION

Individuals with physical disabilities often face significant challenges in performing daily activities that involve operating household electrical appliances requiring manual interaction. People with motor impairments, limited limb movement, or neurological disorders may find it difficult or even impossible to switch on lights, operate fans, or control basic home devices. With advancements in assistive technology, the integration of artificial intelligence and embedded systems has opened new possibilities for developing intelligent, accessible, and user centered solutions. These technologies help enhance independence and improve the overall quality of life for physically challenged individuals. Therefore, designing an intuitive and touch free system for controlling home appliances is an essential step toward creating inclusive smart environments.

Recent developments in computer vision, machine learning, and gesture recognition have enabled the creation of systems capable of accurately interpreting user actions. By integrating these technologies with a Raspberry Pi, it is possible to develop a cost effective and flexible embedded system that supports real-time processing, motion detection, and behavior analysis without requiring expensive hardware. AI based systems allow users to control appliances through natural gestures, facial expressions, and simple body movements, thereby reducing the dependency on physical interaction. This approach eliminates the need for manual switches and voice commands, making the system more suitable for users with mobility or speech limitations.

3.2 PROPOSED SYSTEM

This study analyzes how to enhance existing home automation systems for helping people with disabilities through a new advanced AI based behavioral or motion based control framework that has adaptive learning flexible decision making capabilities to provide real time reliable energy efficient appliance control based on user behavior analysis, motion detection and appliance activity state using sensor fusion techniques. The new framework does not depend on traditional methods of controlling appliances (i.e., predetermined rules) but instead creates a system with adaptive capabilities to accommodate fluctuations in user movements, changing daily patterns, differences in weather or environmental conditions, and abnormal or irregular activities to make sure that each person's experience with automation will be unique, efficient, and appropriate.

The core of the system is a higher order predictive behavior model used with a Sliding Mode Control (SMC) based decision making rule to improve robustness against sensor errors, motion noise, and disturbances from the environment. To address the limitations of abrupt failure through chattering in conventional SMC type methods, the design also includes adaptive control surfaces and smooth transition logic. Adaptive control surfaces support safe and gradual switching of appliances using relay drivers. Devices (e.g., fans) will have a life gain from the stabilization (decreased chattering) and the ability to work reliably throughout

variable user activity conditions. The intelligent co ordination created between the real time detection of motion and the use of long term behaviour prediction is a vital element of the proposed system. Motion sensors monitor current activity performed by the user. Predicted algorithms aid in determining ideal times to use each appliance based on historical CSV logs of usage patterns.

The combination of instantaneous states of motion detection with anticipated frequency of use establishes conditions whereby an appliance will be engaged when there are a match of the behaviour pattern and the presence detection. This reduces instances of appliances turning on unnecessarily, reduces wasted energy and operates the appliances in accordance with actual user requirement. In addition, logic used for disturbance estimation will allow for inaccuracies related to both the noise level of the sensors, behind the scenes, or inconsistencies in the presence patterns, to be compensated, thereby achieving a higher degree of reliability.

The proposed system's design takes into account a smooth transition from predicted activation time periods (when appliances are on), separately identified activation event states (via motion sensors), and periods when appliances are off (energy-saving mode). Sensor data, along with timestamps and learned user behavior history, allow the user to select the most efficient and appropriate method for appliance operation at any given time, while still providing the user with comfort and safety via energy savings. Based on experimental test results, as well as simulation testing, it has been determined that the new AI based system outperforms current rule based, timer based, and simple motion based automation systems in terms of better speed of response, accuracy of prediction, fewer erroneous activation events, and increased energy efficiency.

Overall, the proposed system establishes a conceptual foundation for next generation assistive smart home environments. By integrating predictive behavioral learning, adaptive control strategies, motion sensor fusion, and disturbance aware decision logic, the model provides a reliable, personalized and Energy efficient automation solution for physically challenged individuals. Such a strategy will also help not only to improve accessibility and independence of usage but also to make appliances safer, decrease wasteful energy use, and open the way to new intelligent assistive technologies of future

3.3 System Requirements

3.3.1 Hardware Components

The proposed system requires several hardware components that work together to capture user activity and control appliances. The Raspberry Pi 3 serves as the central processing unit, responsible for collecting data from sensors, processing it, and executing control actions. The PIR motion sensor is used to detect the presence and movement of the user by sensing infrared radiation emitted by the human body, thereby activating the system only when required. The WCS current sensor is employed to monitor the electrical status of appliances by measuring the current flow, which helps in determining whether an appliance is in an ON or OFF state. The relay module acts as an interface between the low voltage control system and high voltage appliances, enabling safe switching operations. Additionally, a stable power supply is required to ensure continuous and reliable operation of all components in the system.

Raspberry Pi 3

- Acts as the central processing unit
- Collects sensor data and controls appliances

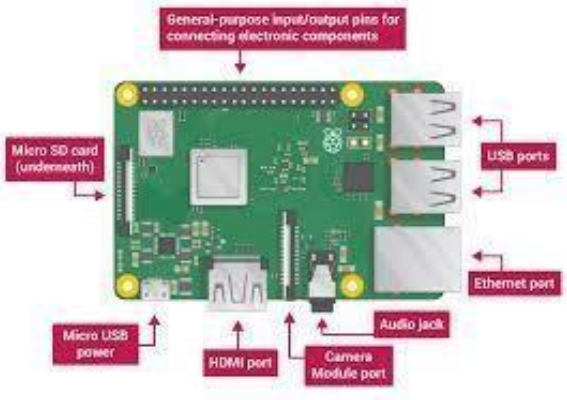

Figure : 3.1 Raspberry Pi 3

PIR Motion Sensor

- Detects user presence and movement
- Activates the system when motion is identified

WCS Current Sensor

- Monitors the electrical status of appliances
- Helps verify whether the appliance is ON or OFF

Relay Module

- Controls switching of electrical appliances
- Provides safe operation between low and high voltage

Power Supply

- Provides stable voltage to the system components

3.3.2 Software Components

The software components of the system play a crucial role in enabling intelligent processing and communication between different modules.

Python

Python is used as the primary programming language to develop the system logic and integrate machine learning functionalities. It processes sensor data and controls the overall system operations. Its simplicity and rich libraries make it suitable for real-time embedded applications.

FlaskAPI

Flask API is used to handle data communication between different modules of the system. It receives sensor data, processes it, and forwards it to the required components. This ensures smooth interaction between hardware and software layers.

Express.js

Express.js is used for managing backend operations and handling system data efficiently. It supports request handling, data routing, and communication between different services. This helps maintain a structured and responsive system environment.

Linear Regression (ML Model)

Linear regression is a type of supervised machine-learning algorithm that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. It assumes that there is a linear relationship between the input and output, meaning the output changes at a constant rate as the input changes. This relationship is represented by a straight line.

For example : To predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied.

Here

Independent variable (input): Hours studied because it's the factor we control.

Dependent variable (output): Examscore because it depends on.

Formula ,

$$h(x_1,x_2,\ldots,x_k)=\beta_0+\beta_1x_1+\beta_2x_2+\ldots+\beta_kx_k$$

Where:

- $x_1, x_2, \ldots, x_k$ are the independent variables.
- β_0 is the intercept.
- $\beta_1, \beta_2, \ldots, \beta_k$ are the coefficients, representing the influence of each respective independent variable on the predicted output

3.4 Architecture of Assistive Appliance Control System

The architecture of the proposed system is designed in a modular manner to ensure efficient operation and easy integration of components. It consists of three main modules, namely the Motion and Behavior Recognition Module, the AI Based Decision and Control Module, and the Appliance Control and Safety Module. These modules work together in a coordinated manner to detect user activity, process information intelligently, and control appliances safely and effectively.

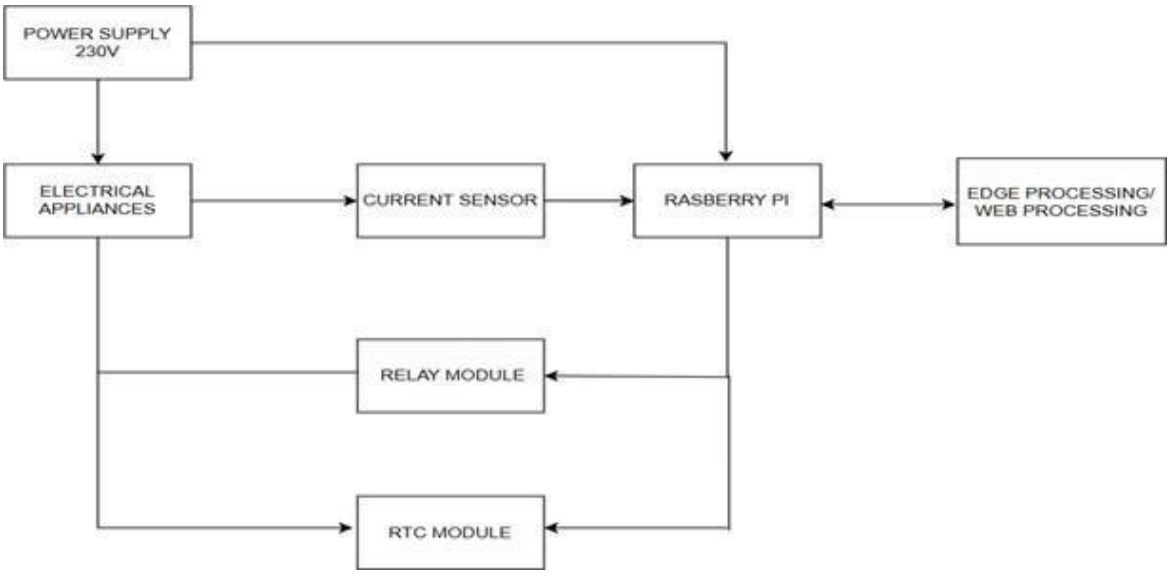

Figure : 3.2 Architecture Diagram

3.4.1 Motion & Behavior Recognition Module

The Motion and Behavior Recognition Module is responsible for detecting user presence and analyzing activity patterns. This module uses the PIR motion sensor to identify movement and determine whether a user is present in the environment. At the same time, the WCS current sensor monitors the status of electrical appliances by measuring current flow. The data collected from these sensors is sent to the Raspberry Pi, where it is

processed and prepared for further analysis. The system also considers time based information to understand user habits and routines. Using the Linear Regression model, the system learns behavioral patterns such as regular usage timings of appliances. This module ensures accurate recognition of user activity and provides meaningful input to the decision-making module, forming the foundation of the entire system.

3.4.2 Appliance Control & Safety Module

The Appliance Control and Safety Module is responsible for executing the final control actions and ensuring safe operation of electrical devices. Once the decision is made, the Raspberry Pi sends control signals to the relay module, which switches the appliances ON or OFF accordingly. This module includes safety mechanisms to prevent incorrect or unsafe operations. It verifies conditions before executing commands to avoid false triggering caused by noise or sensor errors. Additionally, the system monitors the appliance status using the current sensor to ensure that the desired action has been successfully performed. In case of any abnormal condition, such as unexpected switching or sensor mismatch, the system takes corrective action by ignoring invalid inputs or stopping the operation. This module ensures reliability, safety, and efficient functioning of the system. . When motion is detected, the Raspberry Pi collects the sensor data and sends it for processing through the software layer.

3.5 System Flow Diagram

The system flow begins with the initialization of all hardware components, including the Raspberry Pi, PIR motion sensor, and current sensor. The PIR sensor continuously monitors the environment to detect user movement, while the current sensor tracks the status of the connected

appliances. When motion is detected, the Raspberry Pi collects the sensor data and sends it for processing through the software layer.The collected data is analyzed using the Linear Regression model, which predicts user behavior based on previously recorded patterns. The system then combines real time sensor input with prediction results to make a decision regarding appliance control. If the conditions are satisfied, a control signal is sent to the relay module to switch the appliance ON or OFF.

This process runs continuously in a loop, allowing the system to respond dynamically to user activity while ensuring efficient and accurate operation of electrical appliances.

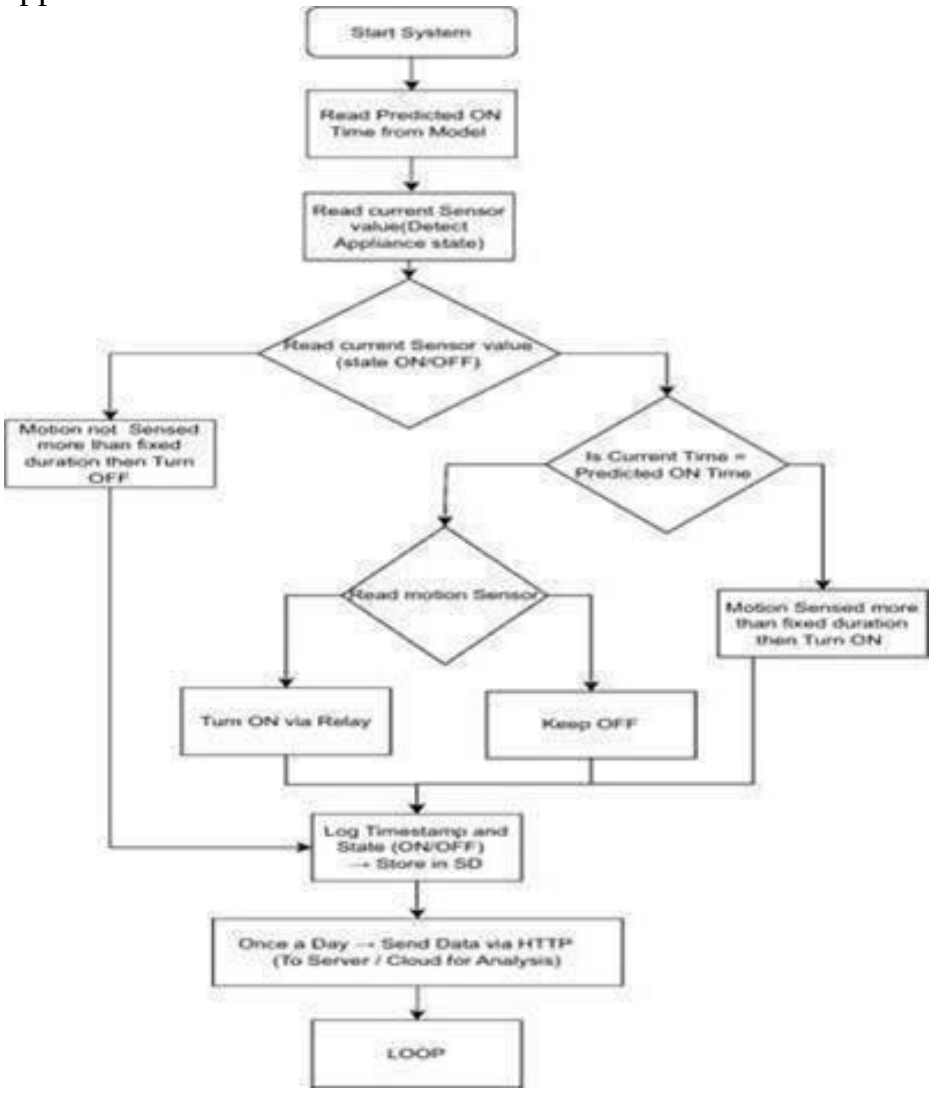

Figure : 3.3 System Flow Diagram

3.6 Summary

The proposed system provides an effective and intelligent solution for assisting physically challenged individuals in controlling household electrical appliances. By combining motion detection, behavioral analysis, and machine learning, the system enables automatic and contactless operation of devices. The modular architecture ensures efficient functioning, while the integration of safety mechanisms enhances reliability. The use of prediction models improves system adaptability and reduces unnecessary energy consumption. Overall, the system offers a practical and user centered approach to smart appliance control, contributing to improved independence, comfort, and quality of life.

CHAPTER 4

MOTION & BEHAVIOR RECOGNITION UNIT

4.1 Introduction to Motion & Behavior Recognition Module

The Motion and Behavior Recognition Module is a fundamental part of the proposed system, responsible for detecting user presence and understanding activity patterns. This module captures real time input from sensors and converts it into meaningful data that can be used for intelligent decision making. It enables the system to recognize whether a user is present and how they interact with the environment. By combining motion sensing and behavior analysis, the system ensures accurate and efficient operation of electrical appliances without requiring physical interaction. This module plays a crucial role in improving accessibility and providing a user-friendly experience for physically challenged individuals.

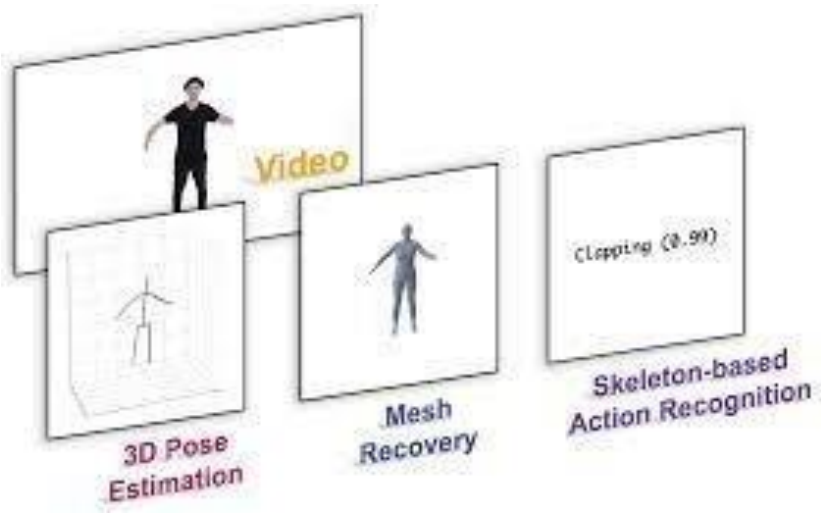

Figure : 4.1 Motion & Behavior Recognition Diagram

4.2 Types of Inputs in Recognition System

The recognition system relies on multiple types of inputs to accurately understand user activity. These inputs help the system make better decisions and reduce errors.

4.2.1 Motion Detection Input

Motion detection is achieved using a PIR sensor, which identifies the presence of a user by detecting infrared radiation emitted from the human body. When movement is detected, the sensor sends a signal to the Raspberry Pi, indicating user activity. This helps the system activate only when necessary.

4.2.2 Appliance Status Input

The WCS current sensor provides information about the operational status of electrical appliances by measuring current flow. It helps determine whether an appliance is ON or OFF and provides feedback to the system for validation.

4.2.3 Time-Based Input

Time plays an important role in understanding user behavior. By using time-related data, the system can identify patterns such as regular appliance usage at specific times of the day, improving prediction accuracy**.**

4.2.4 Behavioral Pattern Input

Behavioral input is derived from historical data collected over time. The system analyzes repeated user actions and learns patterns, which are later used to predict future appliance usage.

4.2.5 Environmental Input

Environmental factors such as temperature variations, lighting conditions, and background movement can affect sensor readings. The system considers these factors to avoid incorrect detection.

4.3 Components of Motion & Behavior Recognition Module

This module consists of several components that work together to capture and process user activity. The PIR sensor detects motion, while the WCS current sensor monitors appliance status. The Raspberry Pi acts as the central unit that collects and processes data from these sensors. The Flask API is used to handle data processing and communication, while the Linear Regression model is used to analyze behavioral patterns. These components together form an efficient recognition system.

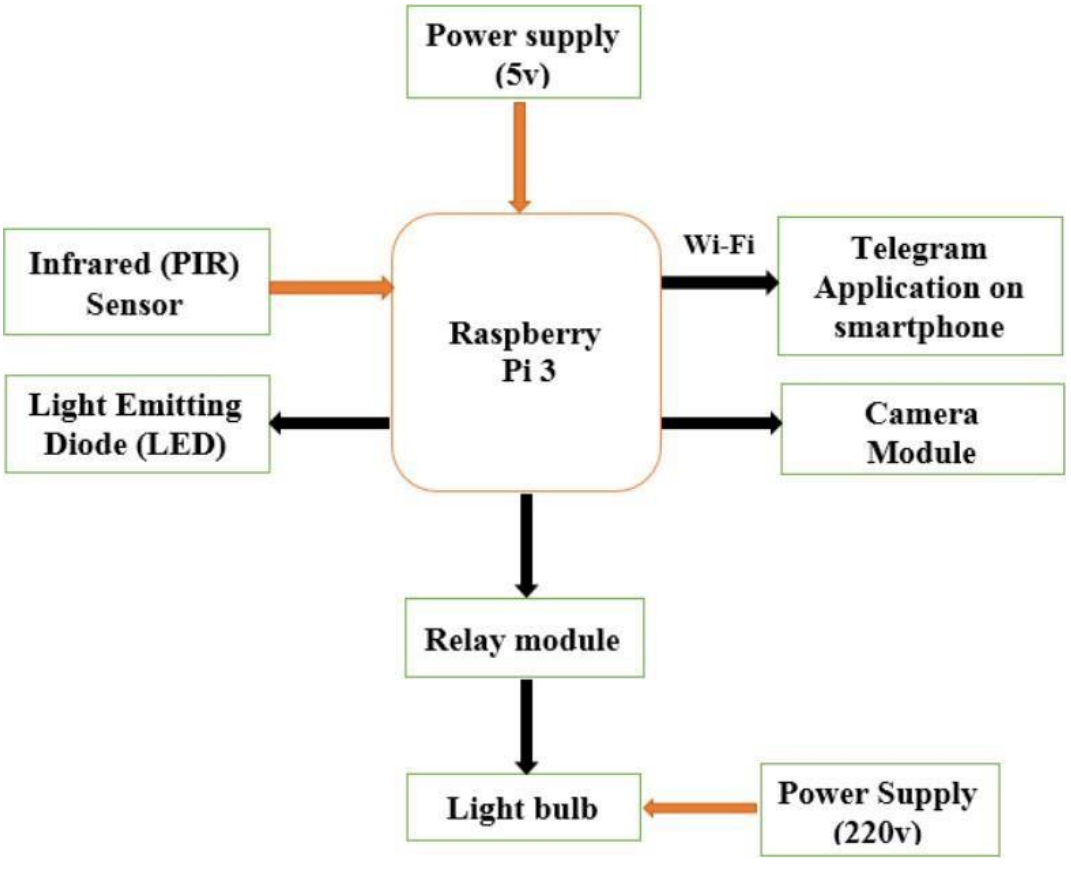

Figure : 4.2 Components of Motion & Behavior Recognition Diagram

4.4 Recognition Methodology

The system follows a structured methodology to recognize user activity. Initially, sensor data is continuously collected from the PIR and current sensors. This data is then processed and cleaned to remove noise and inconsistencies. The system analyzes the processed data along with time based information to identify patterns. The Linear Regression model is applied to learn user behavior and predict appliance usage. Finally, the recognized output is forwarded to the decision module for further action. This step by step approach ensures accurate recognition and reliable system performance.

4.4.1 Block Diagram Motion & Behavior Recognition Unit

The Motion and Behavior Recognition Unit is responsible for collecting and analyzing user related inputs to identify activity and usage patterns. The system begins with multiple input sources, including the PIR motion sensor, current sensor, time based data, behavioral history, and environmental conditions.

The PIR sensor detects user presence by sensing motion, while the current sensor monitors the status of appliances. Time based input helps identify regular usage patterns, and behavioral data is obtained from previously stored records. Environmental factors such as lighting and noise are also considered to improve accuracy.

All these inputs are sent to the Raspberry Pi, which acts as the central processing unit. The data is then processed using the Flask API, where it is cleaned, structured, and prepared for analysis. The processed data is passed to the Linear Regression model, which analyzes patterns and predicts user behavior.

The final output of this module is a recognized activity or predicted usage pattern, which is forwarded to the decision and control module. This ensures that the system responds accurately to user actions and operates appliances efficiently.

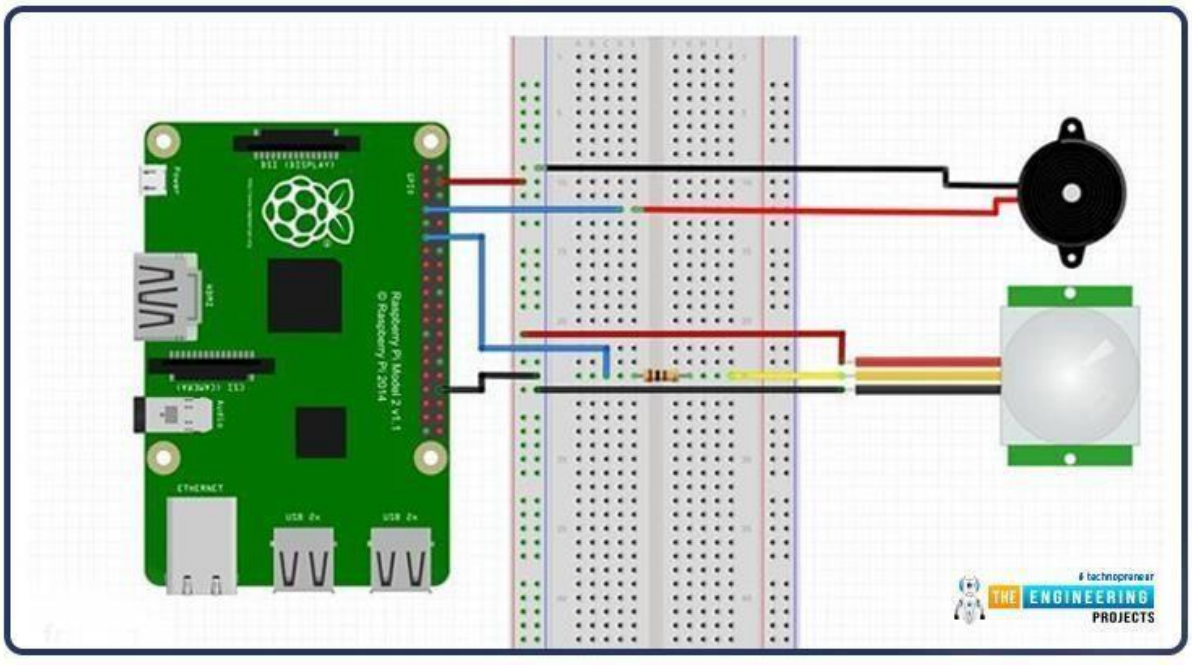

Figure : 4.3 Block Diagram Motion & Behavior Recognition Unit

4.5 Data Processing and Communication

The collected data is processed using the Flask API, which acts as an interface between hardware and software components. It organizes sensor data into a structured format and sends it to the backend system for analysis. Express.js manages data handling and ensures smooth communication between different parts of the system. This process enables real-time data transfer and efficient system operation.

4.6 Challenges in Recognition Module

The recognition module faces several challenges that can affect system performance. Variations in user movement, especially slow or limited motion, can make detection difficult. Environmental noise and external disturbances may lead to false triggering. Sensor inaccuracies and hardware limitations can also impact reliability. Additionally, processing rea time data efficiently on an embedded system like Raspberry Pi requires careful optimization.

4.7 Advantages of Motion & Behavior Recognition

The use of motion and behavior recognition provides several advantages. It enables contactless interaction, making the system suitable for physically challenged individuals. It improves accuracy by combining real time sensing with behavioral learning. The system reduces dependency on manual control and enhances user convenience. It also helps in optimizing energy usage by ensuring appliances operate only when required.

4.8 Future Enhancements

The recognition module can be further improved by integrating advanced machine learning models for better prediction accuracy. Additional sensors can be included to enhance detection capabilities. The system can also be upgraded to support adaptive learning, allowing it to adjust automatically to changes in user behavior. Improvements in processing speed and system efficiency can make the system more robust and scalable.

CHAPTER 5

HARDWARE MODULE FOR MOTION BASED ASSISTIVE CONTROL SYSTEM

5.1 Introduction

The hardware implementation of the proposed AI driven behavioral and motion-based appliance control system focuses on integrating sensing, processing, and control components to achieve reliable and real time operation. The system is designed to assist physically challenged individuals by enabling automatic control of household appliances without manual interaction. It utilizes a Raspberry Pi as the central controller, along with motion and current sensors to capture user activity and appliance status. The collected data is processed and used to make intelligent decisions for appliance control. The hardware setup is designed to be cost effective, energy efficient, and easy to deploy in real world environments. Special attention is given to safety, accuracy, and continuous operation to ensure the system performs reliably under different conditions.

5.2 Controller Selection (RASPBERRY PI 3)

The Raspberry Pi 3 is chosen as the core controller of the system due to its powerful processing capability, compact size, and flexibility in interfacing with various hardware components. It is equipped with a quad core processor, sufficient RAM, built in Wi-Fi, and multiple GPIO pins, making it suitable for real time data processing and control applications. The Raspberry Pi serves as the central unit that collects input from sensors, processes the data using software algorithms, and sends output signals to control appliances. It supports programming languages such as Python, which allows easy integration of machine learning models like Linear Regression. Addition to the Raspberry Pi can handle multiple tasks simultaneously, such as sensor monitoring, data processing, and communication, making it ideal for embedded assistive systems**.**

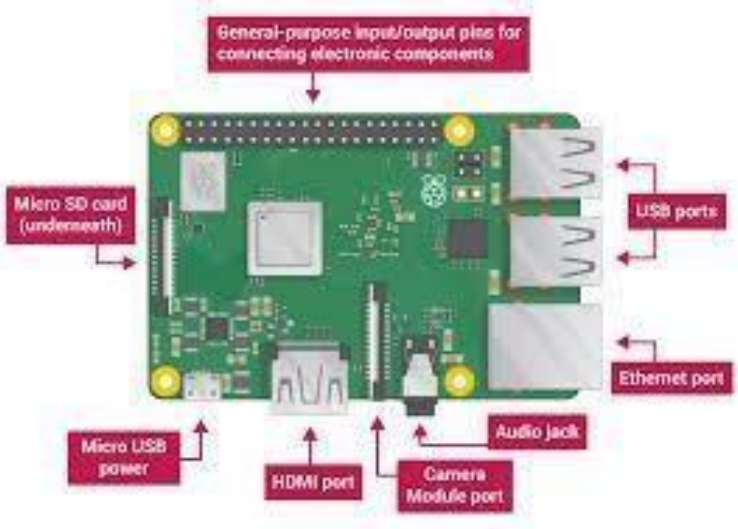

Figure : 5.1 Raspberry Pi 3

5.3 Sensor Integration

Sensor integration is a crucial part of the system, as it enables the detection of user activity and monitoring of appliance conditions. The system uses sensors to gather real time data, which is then processed by the Raspberry Pi to make decisions. Proper integration ensures accurate data collection, minimal delay, and reliable system performance.

5.3.1 PIR Sensor for Motion Detection

The PIR (Passive Infrared) sensor is used to detect human presence and movement. It works by sensing changes in infrared radiation emitted by the human body. When a person enters the detection range, the sensor generates a signal that is sent to the Raspberry Pi. This signal indicates that user activity is present, allowing the system to activate further processing. The PIR sensor helps reduce unnecessary power consumption by ensuring that appliances are operated only when the user is present. It is highly efficient, low cost, and suitable for real time motion detection in indoor environments.

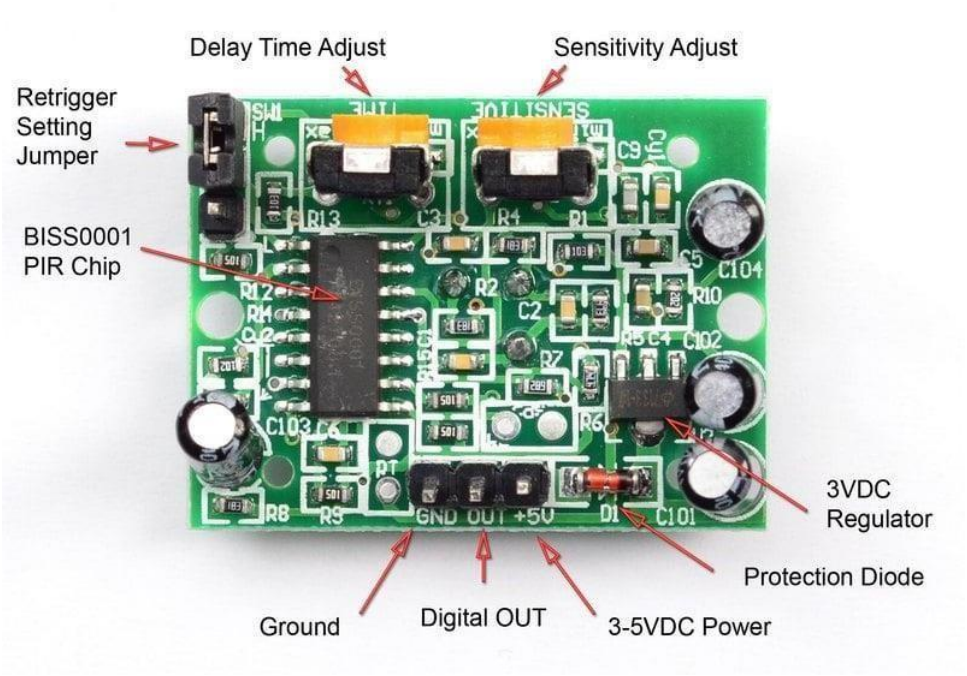

Figure : 5.2 PIR Sensor

5.3.2 WCS Current Sensor for Appliance Monitoring

The WCS current sensor is used to measure the current flowing through electrical appliances. It provides information about whether an appliance is ON or OFF and helps verify if the control action has been successfully executed. This sensor plays a key role in feedback and validation, ensuring that the system operates accurately. By continuously monitoring current flow, the system can also detect abnormal conditions such as unexpected power usage or appliance malfunction. The use of a current sensor improves system reliability and supports fault detection mechanisms.

5.3.3 Sensor Data Integration

The data collected from the PIR sensor and current sensor is integrated within the Raspberry Pi for further processing. The system combines motion detection data

with appliance status information to make accurate decisions. For example, if motion is detected and the appliance is OFF, the system may decide to turn it ON. Similarly, if no motion is detected for a certain period, the system may turn the appliance OFF. This integration ensures that the system responds intelligently to real time conditions and user behavior.

5.4 Relay and Driver Circuit

The relay module is an essential component used to control electrical appliances. It acts as a switching device that connects or disconnects the appliance from the power supply based on control signals received from the Raspberry Pi. The relay provides electrical isolation between the low voltage control circuit and high voltage appliances, ensuring safe operation. When the Raspberry Pi sends a signal, the relay activates and changes its state, thereby turning the appliance ON or OFF. The driver circuit ensures that sufficient current is provided to operate the relay reliably. This setup allows the system to control multiple appliances safely and efficiently.

5.5 Block Diagram Of Hardware Implementation

The hardware implementation of the proposed system consists of multiple components working together to detect user activity and control electrical appliances. The system begins with the power supply unit, which converts AC input into regulated DC voltage required for the Raspberry Pi and sensors.

The PIR motion sensor detects user presence by sensing infrared radiation and sends signals to the Raspberry Pi. At the same time, the WCS current sensor monitors the electrical status of appliances by measuring current flow. These sensors provide real- time input data to the system.

The Raspberry Pi 3 acts as the central controller, where all sensor data is collected and processed.

The relay module acts as a switching device that controls electrical appliances such as lights and fans. It ensures safe operation by isolating the low voltage control circuit from high voltage appliances. The entire system works in a coordinated manner to provide automatic and efficient control of appliances.

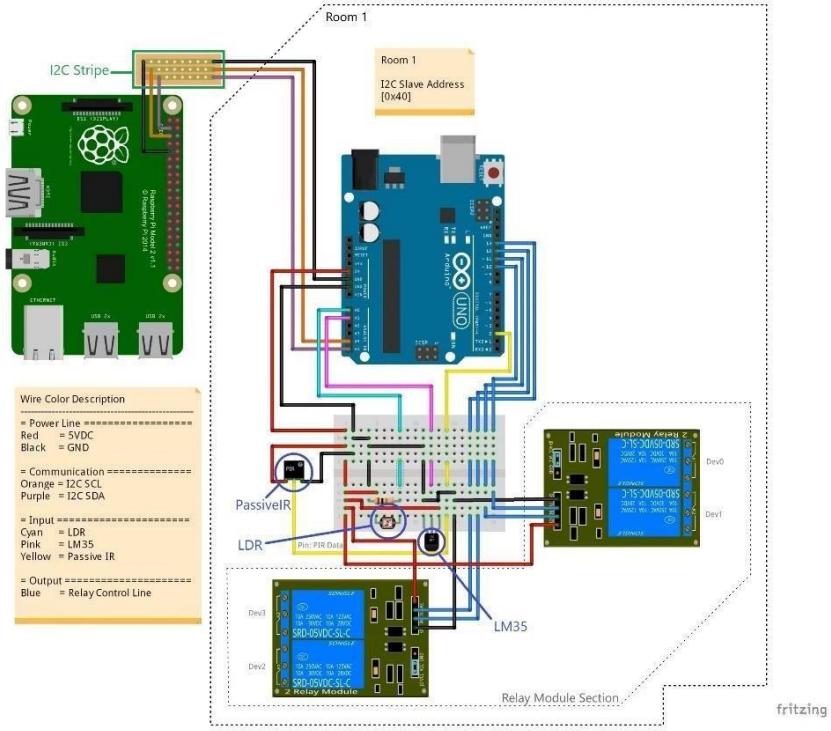

Fig 5.3: Hardware Block Diagram

5.6 Power Supply Design

The power supply design is critical to ensure stable and uninterrupted operation of the system. The Raspberry Pi and sensors require a regulated low voltage DC supply, typically 5V, while the appliances operate on high voltage AC power. A proper power supply unit is used to convert and regulate voltage levels as required. Protection mechanisms such as voltage regulation, grounding, and isolation are implemented to prevent damage to components and ensure user safety. A stable power supply ensures that the system operates continuously without interruptions, which is especially important for assistive applications.

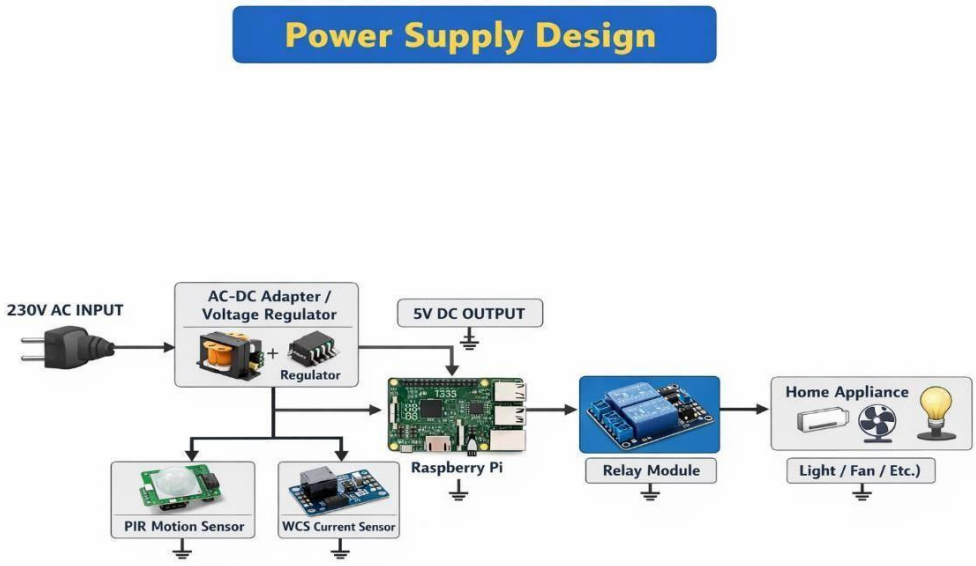

Figure: 5.4 Power Supply Connection

5.7 Communication Modules

Communication between different components of the proposed system is achieved through software based interfaces that ensure smooth and efficient data flow. The Raspberry Pi acts as the central unit that collects data from sensors such as the PIR motion sensor and WCS current sensor. This data is then transmitted to the processing layer using the Flask API, which serves as an interface between the hardware and software components. Flask handles incoming sensor data, performs initial processing, and organizes it into a structured format suitable for further analysis.

Once the data is processed, it is forwarded to the backend system managed by Express.js. The Express.js framework is responsible for handling data management, routing, and coordination between different modules of the system. It ensures that requests and responses are processed efficiently, allowing seamless interaction between the machine learning model, data storage, and control logic. This layered communication structure helps maintain a clear separation between data acquisition, processing, and decision making stages.

The communication setup also supports real time data transfer, enabling the system to respond quickly to changes in user activity. Continuous data exchange ensures that the system remains synchronized and operates without delay. Additionally, this framework allows the system to store logs of user behavior and appliance usage, which can be used for further analysis and improving prediction accuracy.

5.8 Summary

The hardware implementation of the proposed system provides a strong foundation for intelligent appliance control. By integrating the Raspberry Pi, motion sensor, current sensor, and relay module, the system is capable of detecting user activity, processing data, and executing control actions accurately. The design ensures safety through electrical isolation and stable power supply, while the integration of sensors improves reliability and responsiveness. Overall, the hardware setup is efficient, cost effective, and well suited for assisting physically challenged individuals in managing their daily activities independently.

CHAPTER 6

SOFTWARE MODULE FOR AI BEHAVIOR CONTROL SYSTEM

6.1 Introduction

The software implementation of the proposed system plays a vital role in enabling intelligent appliance control through motion and behavioral analysis. It integrates sensor data processing, machine learning, and communication between different system components. The software is designed to process real-time inputs from sensors, analyze user behavior, and generate appropriate control signals for appliances. By using Python, Flask API, and Express.js, the system ensures smooth interaction between hardware and processing layers. The incorporation of a machine learning model further enhances the system's ability to learn user patterns and make accurate predictions, resulting in efficient and automated operation.

6.2 Motion and Behavior Analysis using Machine Learning

The system uses machine learning techniques to analyze user behavior and improve decision making. Instead of relying only on real-time sensor data, the system also considers historical usage patterns to predict appliance requirements. The Linear Regression model is used to identify relationships between variables such as time, motion detection, and appliance usage. This allows the system to estimate the likelihood of appliance activation at a given time. By combining real time input with learned patterns, the system achieves better accuracy and reduces unnecessary switching of appliances.

6.2.1 Linear Regression Model Implementation

The Linear Regression model is implemented using Python libraries to analyze historical data collected from the system. The model takes input features such as time of day, motion detection status, and previous appliance usage and

predictions about future appliance requirements. The model is trained using stored data and continuously updated as new data becomes available. Due to its simplicity and low computational requirements, Linear Regression is suitable for execution on the Raspberry Pi. It provides fast predictions and supports real time, decision making without introducing significant processing delays.

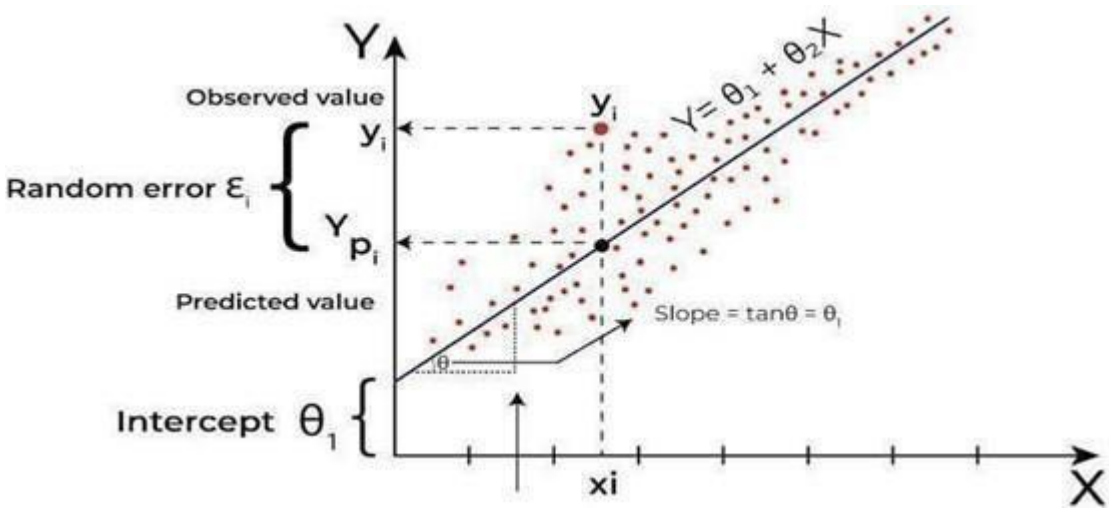

Figure : 6.1 Linear Regression Model

Here

Independent variable (input): Hours studied because it's the factor we control.

Dependent variable (output): Examscore because it depends on.

Formula ,

$$h(x_1,x_2,\ldots,x_k)=\beta_0+\beta_1x_1+\beta_2x_2+\ldots+\beta_kx_k$$

Where:

- $x_1, x_2, \ldots, x_k$ are the independent variables.
- β_0 is the intercept.

- $\beta_1, \beta_2, \ldots, \beta_k$ are the coefficients, representing the influence of each respective independent variable on the predicted output

6.2.2 Data Processing using Flask API

Flask API is used to manage the flow of data between hardware and software components. It receives sensor data from the Raspberry Pi, processes it, and converts it into a structured format for analysis. Flask acts as an intermediate layer that ensures smooth communication between the machine learning model and the control logic. It also handles requests and responses efficiently, allowing real time data processing and system updates. This makes the system more organized and scalable.

6.2.3 Block Diagram Of Software Implementation

The software implementation of the proposed system focuses on processing sensor data, analyzing user behavior, and generating control decisions. The process begins with data input from sensors, including motion detection and appliance status. This data is collected by the Raspberry Pi and sent to the software layer for processing.

The Flask API acts as an interface that receives sensor data, organizes it, and prepares it for further analysis. The processed data is then passed to the Linear Regression model, which analyzes historical data and predicts appliance usage based on user behavior patterns.

The prediction results, along with real time data, are forwarded to the control logic module, where decisions are made regarding appliance operation. The Express.js backend manages data handling and stores information in MongoDB, enabling continuous learning and system improvement.

Finally, the system generates output signals that are sent to the hardware layer, where the relay module controls the electrical appliances. This entire process ensures real-time operation, accurate prediction, and efficient appliance control.

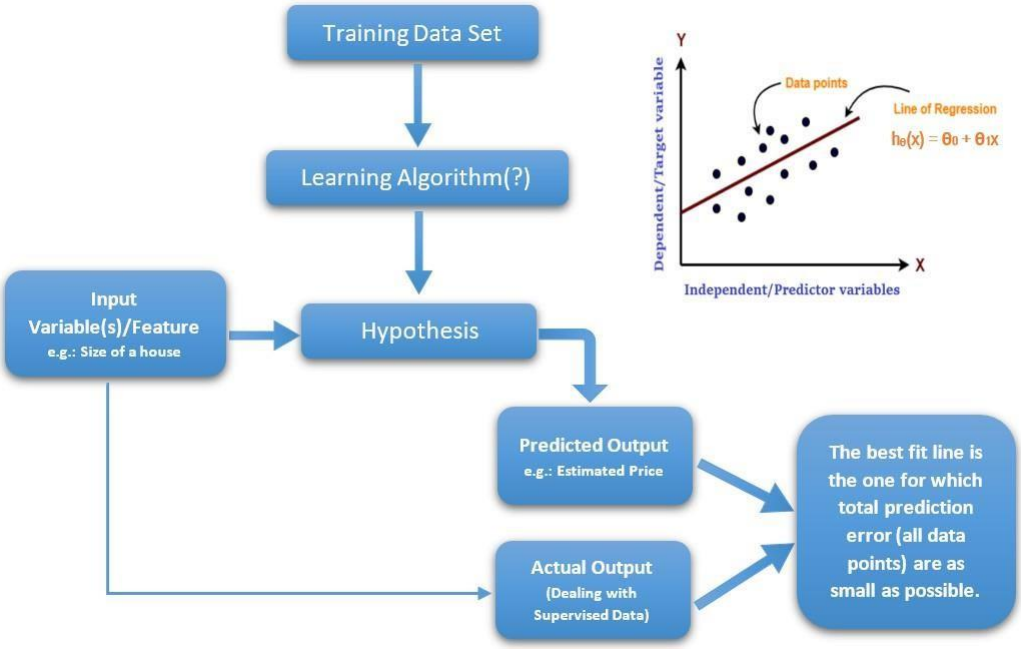

Figure : 6.2 Block Diagram Of Software Implementation

6.3 Dataset and Training Process

The dataset used in the system is generated from real time sensor readings and stored as logs. It includes information such as motion detection status, appliance ON/OFF state, and timestamps. This data is collected over a period of time to capture user behavior patterns. Before training, the data is preprocessed to remove noise, handle missing values, and normalize inputs. The cleaned dataset is then used to train the Linear Regression model. As more data is collected, the model is retrained to improve accuracy and adapt to changes in user behavior.

6.4 Real Time Processing on Embedded Device

The system is designed to perform real time processing on the Raspberry Pi. Sensor data is continuously monitored and processed without delay, ensuring quick response to user actions. The machine learning model runs on the device and provides instant predictions based on incoming data. This edge level processing eliminates the need for external computing resources .

6.5 Integration with Appliance Control Logic

The software module is tightly integrated with the appliance control system. The output generated by the machine learning model and sensor analysis is used to control the relay module. The system combines motion detection, appliance status, and prediction results to make accurate decisions. For example, if motion is detected and the predicted usage condition is satisfied, the system turns the appliance ON. Similarly, if no motion is detected, the system switches the appliance OFF after a certain period. This integration ensures efficient operation, reduces energy consumption, and enhances user convenience.

6.6 Summary

The software implementation provides the intelligence required for the proposed system to function effectively. By integrating machine learning, real time data processing, and efficient communication mechanisms, the system is able to analyze user behavior and control appliances automatically. The use of lightweight algorithms and embedded processing ensures fast and reliable performance. Overall, the software design enhances system adaptability, accuracy, and efficiency, making it suitable for assistive applications aimed at improving the quality of life for physically challenged individuals.

CHAPTER 7

INTEGRATED EDGE PROCESSING AND CONTROL SYSTEM

7.1 Introduction

The hardware and software connectivity of the proposed system plays a crucial role in enabling seamless communication between sensors, processing units, and control mechanisms. The system integrates hardware components such as the Raspberry Pi, PIR sensor, current sensor, and relay module with software technologies including Flask API, Express.js, and MongoDB. This integration ensures efficient data transfer, real time processing, and intelligent decision making. The connectivity framework allows sensor data to be captured, processed, stored, and utilized for machine learning and appliance control. A well designed communication system improves system reliability, scalability, and overall performance.

7.2 Edge Based Intelligent System Architecture

The system architecture is designed to establish a smooth interaction between hardware and software layers. The Raspberry Pi acts as the central node that collects data from sensors and communicates with the software layer. Sensor data such as motion detection and current readings are processed locally and then sent to the backend through Flask API. The backend system, implemented using Express.js, manages the data flow and ensures proper coordination between modules. MongoDB is used as the database to store sensor logs, appliance status, and user behavior data. This layered architecture ensures efficient data handling and supports real-time system operation.

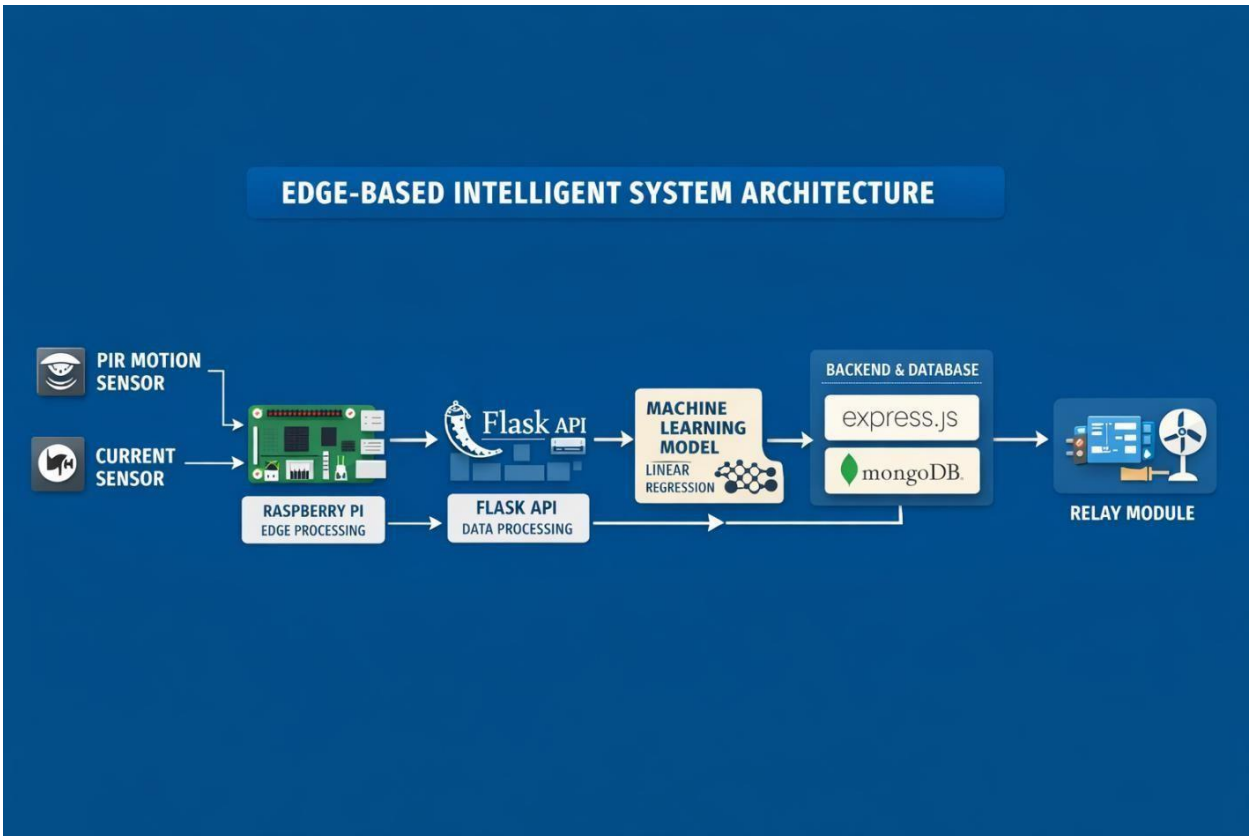

Figure :7.1 System Architecture for Edge Based

7.3 Database Integration using MongoDB

MongoDB is used as the primary database for storing and managing system data. It stores information such as motion detection events, appliance ON/OFF status, timestamps, and prediction results. As a NoSQL database, MongoDB provides flexibility in handling large volumes of unstructured data generated by sensors. The database supports real-time updates and allows easy retrieval of data for analysis and machine learning. By maintaining historical records, MongoDB enables the system to learn user behavior patterns and improve prediction accuracy over time. Its cloud based accessibility also allows future enhancements such as remote monitoring.

7.4 Sensor Data Transmission and Processing

Sensor data transmission begins with the PIR sensor detecting motion and the current sensor measuring appliance status. This data is collected by the Raspberry Pi and sent to the Flask API for processing. Flask acts as an interface that formats and forwards the data to the backend system. The processed data is then stored in MongoDB for future reference. The system ensures that data transmission occurs in real time, enabling quick response to user activity. Efficient data processing reduces delays and improves system performance.

7.5 Behavior Analysis and System Analytics

The stored data in MongoDB is used for analyzing user behavior and system performance. The Linear Regression model utilizes this data to identify usage patterns and predict future appliance requirements. The system analyzes parameters such as time of usage, motion frequency, and appliance activity. This analysis helps in optimizing system performance, reducing energy consumption, and improving prediction accuracy. It also supports fault detection by identifying unusual patterns in appliance behavior.

7.6 Monitoring and Control Interface

The system provides a monitoring and control mechanism through the backend managed by Express.js. It allows tracking of sensor data, appliance status, and prediction results. The stored data can be accessed for analysis and debugging purposes. The system can also be extended to include a user interface for real time monitoring and control. This improves usability and allows caregivers or users to understand system behavior more effectively.

7.7 Summary

The hardware and software connectivity of the proposed system ensures efficient communication and coordination between all components. By integrating Raspberry Pi, sensors, Flask API, Express.js, and MongoDB, the system achieves real time data processing, storage, and intelligent decision-making. The use of MongoDB enhances data management and supports machine learning operations. Overall, the connectivity framework improves system reliability, scalability, and performance, making it suitable for assistive automation applications.

CHAPTER 8
RESULT AND DISCUSSION

8.1 Introduction

This chapter presents the results and analysis of the proposed AI-driven behavioral and motion-based control system for electrical appliances. The system was implemented using Raspberry Pi, PIR motion sensor, WCS current sensor, relay module, and software technologies such as Flask API, Express.js, MongoDB, and a Linear Regression model. The objective of this chapter is to evaluate how effectively the system detects user activity, analyzes behavior, and controls appliances automatically. The results highlight the system's performance in terms of accuracy, response time, reliability, and energy efficiency, demonstrating its suitability for assisting physically challenged individuals.

8.2 Experimental Setup

The experimental setup was designed to test the system in a real-time indoor environment. The Raspberry Pi was connected with a PIR sensor to detect user motion and a current sensor to monitor appliance status. A relay module was used to control devices such as lights and fans. The sensors were placed in suitable positions to ensure accurate detection of user presence and appliance activity.

On the software side, Flask API was used to handle data processing, while Express.js managed backend operations. MongoDB was used to store sensor data, user activity logs, and prediction results. The Linear Regression model was trained using collected data to learn user behavior patterns. Various test scenarios were performed , including user entry and exit, continuous presence, and repeated usage patterns at specific times, to evaluate the system's performance.

8.3 Performance Metrics

The performance of the system was evaluated using multiple metrics, including motion detection accuracy, behavior prediction accuracy, response time, and energy efficiency. These metrics help in understanding the effectiveness and reliability of the system.

8.3.1 Motion Detection Accuracy

The motion detection accuracy depends on the PIR sensor's ability to correctly identify user presence. The system demonstrated a high accuracy of approximately 92% to 95% in detecting motion, with very few false triggers. Proper placement and calibration of the sensor improved detection performance. The system was able to distinguish effectively between user presence and absence, ensuring that appliances were activated only when required.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP → True Positive (correct motion detection)
- TN → True Negative (correct no-motion detection)
- FP → False Positive (wrong detection)

- FN → False Negative (missed detection)

8.3.2 Behavior Prediction Accuracy

The Linear Regression model was used to predict appliance usage based on historical data stored in MongoDB. The model achieved an accuracy of around 85% to 90% in identifying usage patterns such as regular appliance operation at specific times. As more data was collected, the prediction accuracy improved, demonstrating the system's ability to learn and adapt to user behavior over time. This reduced unnecessary appliance switching and improved overall system efficiency.

8.3.3 Response Time Analysis

The response time of the system refers to how quickly it reacts after detecting user activity. The system showed a fast response time of approximately 1 to 2 seconds, due to local processing on the Raspberry Pi. Once motion was detected, the system processed the data and activated the appliance with minimal delay. Efficient communication through Flask API and Express.js contributed to smooth data flow and quick system response**.**

Formula,

Response Time = Output Time – Input time

8.3.4 Energy Efficiency Analysis

The system achieved improved energy efficiency by automatically turning OFF appliances when no motion was detected. It was observed that the system reduced energy consumption by approximately 20% to 30% compared to manual operation. By combining motion detection and behavior prediction.

8.4 System Output

The output of the proposed system is observed through both hardware operation and software interface results. The system successfully demonstrates automatic control of electrical appliances based on user activity and behavior prediction.

On the hardware side, the PIR motion sensor detects user presence and sends signals to the Raspberry Pi. Based on the processed data and prediction results, the Raspberry Pi activates the relay module, which controls the electrical appliance. When motion is detected, the appliance is turned ON, and when no motion is detected for a specific duration, the appliance is turned OFF. The WCS current sensor continuously monitors the appliance status and provides feedback to ensure correct operation.

On the software side, the system provides a dashboard interface that displays user information, appliance activity, and prediction results. The dashboard shows details such as user name, role, and connected appliances. It also includes modules for viewing daily usage data and predicted usage patterns.

The “Daily Usages” section displays recorded data such as date, time, and appliance status (ON/OFF). This data is stored in MongoDB and reflects real time system activity. The “Predict Usages” section shows the output of the Linear Regression model, which predicts appliance usage based on historical patterns.

The system also provides control and update options through the interface, allowing users to monitor and manage appliance activity effectively. The combination of hardware automation and software visualization ensures accurate, efficient, and user friendly system performance.

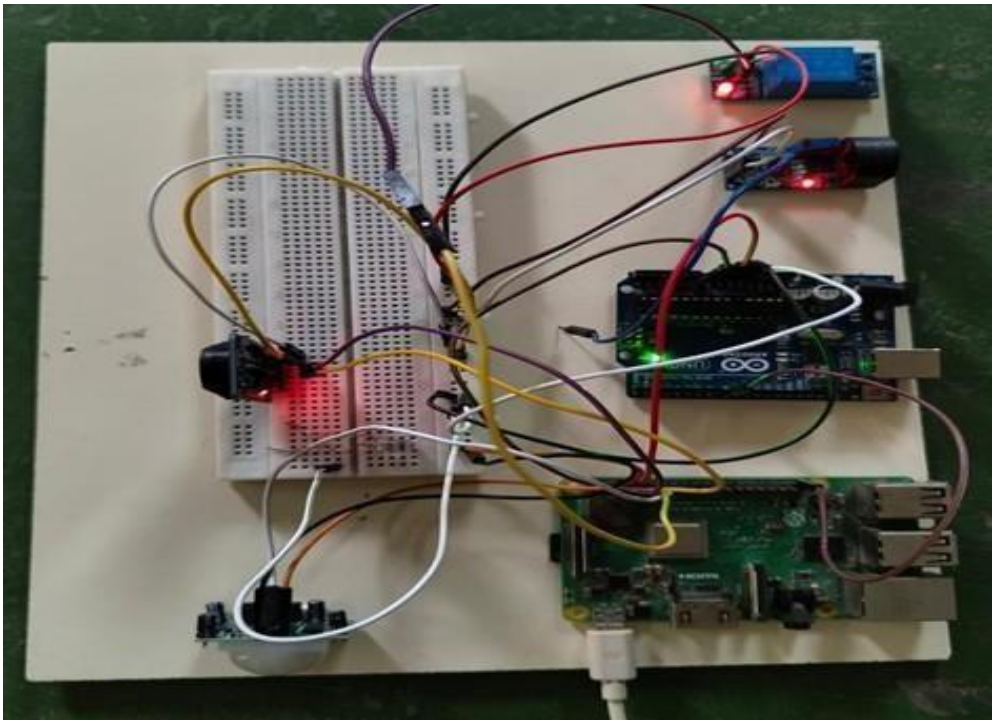

Figure : 8.1 Hardware Output - 1

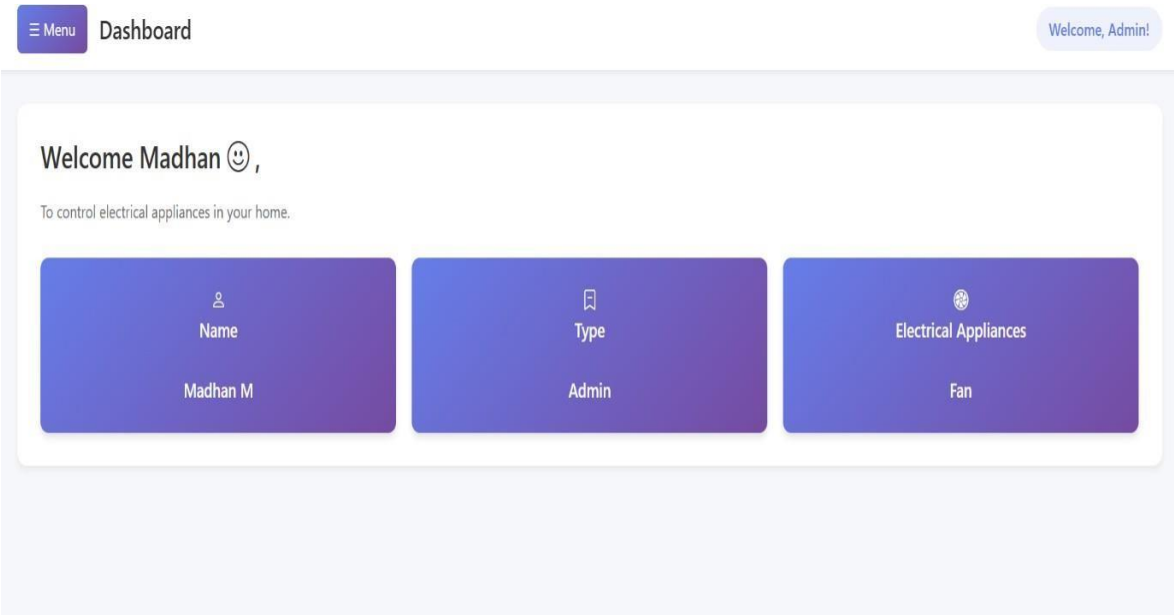

Figure : 8.2 Software Output-1

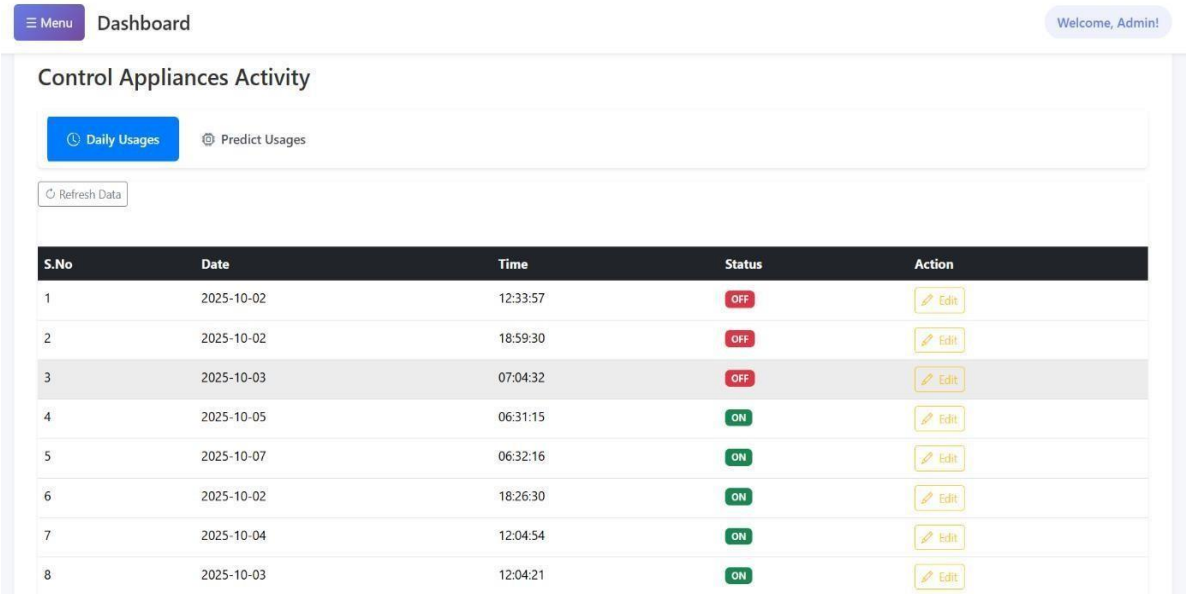

Figure : 8.3 Software Output-2

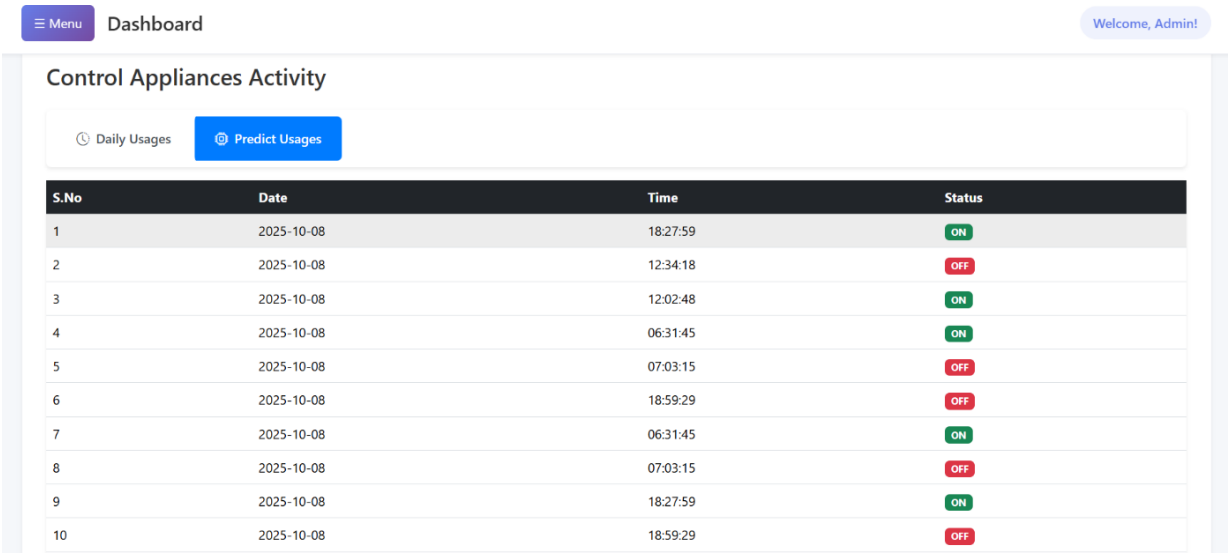

Figure : 8.4 Software Output-3

8.5 Comparsion of Existing System VS Proposed System

The performance of the proposed AI driven behavioral and motion based appliance control system is compared with existing systems to evaluate its effectiveness. Traditional systems such as manual switch based control and basic motion-based automation have several limitations in terms of usability, intelligence, and efficiency. The proposed system overcomes these limitations by integrating machine learning and real time processing. In manual systems, users are required to physically operate switches, which is difficult for physically challenged individuals. Basic motion based systems improve automation but lack intelligence, as they operate appliances only based on motion detection without considering user behavior. In contrast, the proposed system uses a Linear Regression model to analyze historical data and predict appliance usage, making it more intelligent and adaptive.

Table 1 : Comparsion Of Existing System Vs Proposed System

Feature	Existing System	Proposed System
Control Method	Manual switches / Basic motion	Motion + AI-based prediction
User Interaction	Physical / Limited automation	Fully contactless and intelligent
Intelligence	No learning capability	Learns user behavior (ML)
Accuracy	Moderate	High (92%–95%)
Prediction	Not available	Available (85%–90%)
Response Time	Slower / Manual delay	Fast (1–2 seconds)

8.6 Summary

The results and discussion clearly demonstrate that the proposed AI driven behavioral and motion based appliance control system performs effectively in real time conditions. The integration of hardware components such as Raspberry Pi, PIR motion sensor, current sensor, and relay module enables accurate detection of user activity and reliable control of electrical appliances. The system successfully identifies user presence and ensures that appliances are operated only when required, thereby improving efficiency and reducing unnecessary usage.

On the software side, the use of Flask API, Express.js, and MongoDB ensures smooth communication, efficient data handling, and proper storage of system logs. The implementation of the Linear Regression model enhances the intelligence of the system by learning user behavior patterns and predicting appliance usage with good accuracy.

This predictive capability allows the system to adapt over time and improve its performance based on historical data.

The experimental results show that the system achieves high motion detection accuracy, fast response time, and noticeable energy savings. The combination of real time processing and behavior prediction minimizes delays and ensures quick system response. The software dashboard further enhances usability by providing clear visualization of appliance activity and prediction results.

Overall, the proposed system offers a reliable, efficient, and user friendly solution for assisting physically challenged individuals in controlling household appliances. It reduces dependency on manual operation, improves convenience, and enhances quality of life. The system also provides a strong foundation for future enhancements, making it a scalable and practical approach for smart assistive environments.

CHAPTER 9
CONCLUSION & FUTURE SCOPE

9.1 Conclusion

The suggested AI based behavioral and movement based home automation system is a smart, active and efficient system that is aimed at assisting physically disadvantaged people in their everyday living space. With the combination of motion sensors, real time current sensing, and behavior learning based on predictive AI plots, the system will provide a correct and personalized control of the appliances in the house, including lights, fans, and electric loads. The Raspberry Pi is a strong processing board, which integrates real time sensing with learned user patterns to come up with reliable ON/OFF decisions even in the face of uncertain/varying user routines.

The combination of prediction-based control and the motion-based verification improves the strength of the system and averts false switching which will allow Appliances to operate smoothly and safely. Continuous refinement of the prediction made by the system possible by the use of daily CSV logs and adaptive learning also means that the system will be able to react to the shift in user behavior, environmental conditions, or irregular movement patterns. This adaptive feature is much more energy efficient and at the same time, it makes the users more comfortable and independent.

The smart integration of sensor information, machine learning, and actuation that uses relays makes sure that the appliance is only turned on when required minimizing unnecessary power usage and maximizing the life of the electrical equipment and switching devices.

9.2 Future Scope

With the establishment and production of automated systems employing artificial intelligence (AI) to manage the daily household operations like switching light on and off and shutting blinds, we establish a solid base to intelligent assistive living solutions in the future. There are numerous upgrades of the system in the future that will make it highly flexible, precise, and overall better. One of the main areas of concern today is the incorporation of more advanced machine learning models that are able to forecast the user patterns over a longer time frame than the existing historical data. Improved neural networks learning methods will also enable us to understand better how to identify and react to user behavior as time goes on as we keep collecting and analyzing motion data and environmental data that influences our day to day activities. This better predictive will make the controller more proactive in the control of energy consumption and will lead to a better support and assistance to individuals with disabilities that need the help with their daily activities.

The other improvement is to increase the sensing environment and incorporate more IoT and smart sensors like temperature sensors, wearable health trackers, or camera based gesture recognition modules as well. Such additions would assist the system in getting to know the user context, health, and the level of comfort to allow a more personalized and adaptive control of the appliances. the electricity usage may be optimized, waste minimized, and the long term sustainability of the home automation grid may be enhanced with the help of the energy monitoring modules and load forecasting algorithms.

Cloud based learning and edge cloud hybrid architecture as well as federated learning can also be used in future versions of the system. The architectures enable continuous updating of models, safe data processing, and optimization in real time, without breaching privacy of the user.

CHAPTER 10
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APPENDIX 1

S.NO	NAME OF THE COMPONENTS	QUANTITY	COST
1.	RASPBERRY PI 3	1	3,600
2.	WCS CURRENT SENSOR	1	200
3.	PIR MOTION SENSOR	1	220
4.	RELAY MODULE	1	200
	TOTAL COST		4220

APPENDIX 2

HARDWARE CODE

```
 import RPi.GPIO as GPIO
import time
from datetime import datetime
import smbus2
import os
# ---------- DS3231 SETUP ----------
I2C_BUS = 1
DS3231_ADDR = 0x68
bus = smbus2.SMBus(I2C_BUS)
def bcd_to_dec(bcd):
    return (bcd >> 4) * 10 + (bcd & 0x0F)
def read_ds3231_time():
    data = bus.read_i2c_block_data(DS3231_ADDR, 0x00, 7)
    return datetime(
        bcd_to_dec(data[6]) + 2000,
        bcd_to_dec(data[5] & 0x1F),
        bcd_to_dec(data[4]),
        bcd_to_dec(data[2] & 0x3F),
        bcd_to_dec(data[1]),
        bcd_to_dec(data[0])
    )
# ---------- GPIO SETUP ----------
GPIO.setmode(GPIO.BCM)
LOG_INPUT_PIN = 17
```

```
PASS_INPUT_PIN = 27
PASS_OUTPUT_PIN = 22
GPIO.setup(LOG_INPUT_PIN, GPIO.IN, pull_up_down=GPIO.PUD_DOWN)
GPIO.setup(PASS_INPUT_PIN, GPIO.IN, pull_up_down=GPIO.PUD_DOWN)
GPIO.setup(PASS_OUTPUT_PIN, GPIO.OUT)
# Initialize output mirror
GPIO.output(PASS_OUTPUT_PIN, GPIO.input(PASS_INPUT_PIN))
last_log_state = GPIO.input(LOG_INPUT_PIN)
# ------------ CSV SETUP ------------
CSV_FILE = "state_log.csv"
# Create file if not exists
if not os.path.exists(CSV_FILE):
        f.write("timestamp,state\n")
print("CSV logging started...")
tr:
    while True:
        # ----  LOGGING (only on change) ----
        current_log_state = GPIO.input(LOG_INPUT_PIN)
        if current_log_state != last_log_state:
            rtc_time = read_ds3231_time()
            timestamp = rtc_time.strftime("%Y-%m-%d %H:%M:%S")
            state = "ON" if current_log_state else "OFF"
            with open(CSV_FILE, "a") as f:
                f.write(f"{timestamp},{state}\n")
            last_log_state = current_log_state
    GPIO.cleanup()
    bus.close()
```

SOFTWARE CODE

```
 const express = require("express");
const mongoose = require("mongoose");
const axios = require("axios");
const cors = require("cors");

const app = express();

const PORT = process.env.PORT || 8000;
const MONGO_URI = process.env.MONGO_URI;
const ML_URL = process.env.ML_URL;

app.use(express.json());
app.use(cors());

app.get("/", (req, res) => {
  res.send("Node Server Running Successfully 🚀");
});

const User = require("./modal/user");
const PredictionResponse = require("./modal/predictionResponseModal");

app.get("/DBdata", async (req, res) => {
  try {
```

```
  const users = await User.find();
   res.json(users);
  }
   catch (error) {
    res.status(500).json({ message: error.message });
  }
});

app.put("/update/:id", async (req, res) => {
const { id } = req.params;
const { State, Timestamp } = req.body;

try {
  const updatedUser = await User.findByIdAndUpdate(
    id,
    { State, Timestamp },
    { returnDocument: 'after' } // Fixes the deprecation warning
  );

  if (!updatedUser) {
    return res.status(404).json({ message: "Record not found" });
  }

  res.json(updatedUser);
```

```
  } catch (error) {
    res.status(500).json({ message: error.message });
  }
});
  app.get("/predictiondb", async (req, res) => {
    try {
      const predictions = await PredictionResponse.find();

      res.json(predictions);
    } catch (error) {
      res.status(500).json({ message: error.message });
    }
  });

app.post("/run-ml", async (req, res) => {
  try {

    // ◈ Get data from MongoDB
    const users = await User.find();

    // ◈ Send data to Flask
    const flaskResponse = await axios.post(
      `${ML_URL}/predict`,
      users,
```

```
    { timeout: 60000 }
  );

  const predictions = flaskResponse.data;

  // ◈ Store predictions
  const savedData = await PredictionResponse.insertMany(predictions);

  res.status(200).json({
    success: true,
    message: "Predictions stored successfully ✅",
    totalStored: savedData.length,
    data: savedData
  });

} catch (error) {
  console.error("Error calling ML service:", error.message);

  res.status(500).json({
    success: false,
    message: "Failed to get ML prediction",
    error: error.message
  });
}
```

```
});

mongoose
  .connect(MONGO_URI)
  .then(() => {
    console.log("MongoDB connected successfully ✅");

    app.listen(PORT, () => {
      console.log(`Server running on port ${PORT} 🚀`);
    });
  })
  .catch((err) => {
    console.log("MongoDB connection error:", err.message);
  });

from flask import Flask, request, jsonify
import joblib
import numpy as np
from datetime import timedelta
import pandas as pd
import os

app = Flask(__name__)
```

```
# 🔥 Load Pickle Files (make sure they are in same folder)
models = joblib.load("ml_models.pkl")
daily_on_times = joblib.load("daily_on_times.pkl")
lookback = joblib.load("lookback.pkl")

@app.route("/predict", methods=["POST"])
def predict():
    try:
        print("📥 Request received from Express")

        # 🧠 Prepare input for prediction
        prediction_input = daily_on_times.iloc[-lookback:].values.flatten().reshape(1, -1)

        # 📅 Predict for next day
        next_day = daily_on_times.index[-1] + timedelta(days=1)

        predicted_on = []

        # 🔥 Predict ON times using ML models
        for model in models:
            mins = model.predict(prediction_input)[0]
            secs = mins * 60

            h = int(secs // 3600)
```

```
    m = int((secs % 3600) // 60)
    s = int(secs % 60)

    predicted_on.append({
        "Timestamp": pd.to_datetime(
            f"{next_day.date()} {h:02d}:{m:02d}:{s:02d}"
        ),
        "State": "ON"
    })

prediction_df = pd.DataFrame(predicted_on)

# 🔥 Generate OFF times using fixed average duration
average_duration_minutes = 31.5  # adjust if needed

off_predictions = []

for i, row in prediction_df.iterrows():
    off_time = row["Timestamp"] + timedelta(
        minutes=average_duration_minutes
    )

    off_predictions.append({
        "Timestamp": off_time,
```

```
            "State": "OFF"
        })

    off_df = pd.DataFrame(off_predictions)

    # 🔄 Combine ON + OFF
    final_prediction = (
        pd.concat([prediction_df, off_df])
        .sort_values("Timestamp")
        .reset_index(drop=True)
    )

    # Convert Timestamp to string for JSON response
    final_prediction["Timestamp"] = final_prediction["Timestamp"].astype(str)

    print("✅ Prediction completed")

    return jsonify(final_prediction.to_dict("records"))

  except Exception as e:
    print("❌ Error:", str(e))
    return jsonify({"error": str(e)}), 500

if __name__== "__main__":
```

```
app.run()
```

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The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

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